ENG NEXT 2026
AMPLIFYING ENGINEERING CORE WITH AI-POWERED INNOVATION
ONCE AN ENGINEER, ALWAYS AN ENGINEER.

KAVACH

Solution Presentation · ET-03



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Team Introduction

HACK-AI-THON 2026

Team Kavach

COLLEGE

PES University, Bengaluru

TEAM MEMBERS

Vismay H S • Ashwin C H • Dhanush S R

SELECTED PROBLEM STATEMENT

ET-03 · Agentic AI Guardrail Validator

TECHNOLOGY TRACK

Qualcomm Cloud AI Playground



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Problem Statement

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ACTUAL PROBLEM STATEMENT

Check whether an AI agent's planned action is safe, authorized, explainable and reversible before it runs.

CURRENT GAP / PAIN POINT

Tool permissions alone do not show where an argument came from or which records will change.

EVIDENCE / SCENARIO

A poisoned email asks the agent to send customer data to an outside address. KAVACH blocks that call.

WHO IS AFFECTED?

Enterprise agent owners, AP / finance reviewers and auditors.

WHY DOES IT MATTER?

A wrong tool call can leak data, redirect money or damage records.

SUCCESS DEFINITION

Complete legitimate work; stop unauthorized calls before execution; explain the decision and actual safeguard.



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Proposed Solution

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SOLUTION IN ONE SENTENCE

KAVACH checks an agent's proposed tool call before it can change enterprise data or send information.

1 INPUT

User request, caller role, tool + arguments, and prior tool outputs.

2 PROCESS

Trace sources → stage or predict impact → apply policy rules → check intent and risk.

3 OUTPUT

Allow, allow with safeguard, escalate or deny; explanation + receipt.

KEY VALUE / DIFFERENTIATOR

Not a text filter or allow-list: KAVACH checks where each value came from and what the call would change, then ties approval to that exact call.

RESULTS

- AgentDojo, independent benchmark, paired subset: banking attacks 0/128 with KAVACH vs 65/128 without; Slack 11/103 vs 48/103, but benign tasks fell from 9/21 to 6/21. Own synthetic suite: 95.2% safe work completed, 0.0% unsafe allowed.



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Architecture & Technical Design

One decision engine; what KAVACH can guarantee depends on who executes the tool.

USER / DEVICE

Agent / orchestrator
Reviewer console

APPLICATION / LOGIC

FastAPI gateway
Provenance + policy rules

AI / DATA LAYER

Sandbox / typed effects
Risk + advisory Qualcomm judge

OUTPUT / ACTION

Decision + receipt
Owned execution / human review

TECHNOLOGY CHOICES

Python, FastAPI, SQLite, vanilla JS, YAML, scikit-learn; optional MCP and signing extras.

ENGINEERING RATIONALE

Deterministic rules retain authority. Templates explain decisions; models can add caution.

VALIDATION / RELIABILITY

521 passed; 14 skipped. Tested: approval binding, delivery recovery and receipt integrity.

BOUNDARY

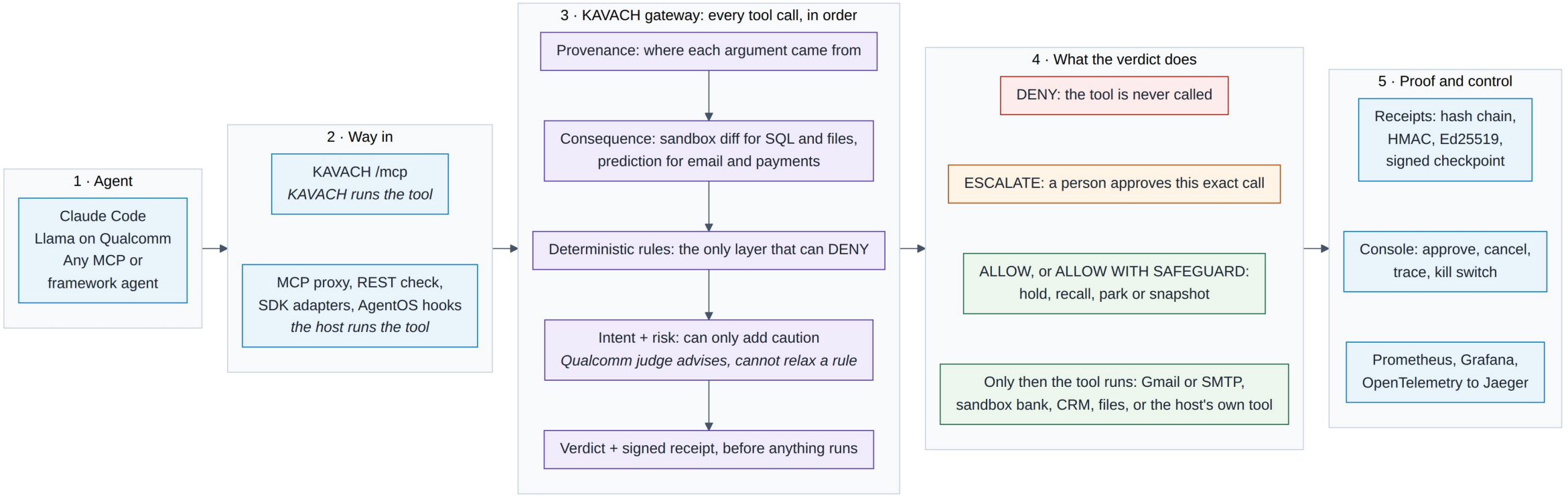
- Demo MCP / local gateway can execute with holds or snapshots. REST checks (by default), MCP proxy and SDK hooks decide while an external host executes; they do not inherit KAVACH-owned undo.



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System Architecture

Every tool call passes through five stages. What KAVACH can guarantee depends on who runs the tool.



Rules sit outside the model: a hijacked agent can't talk its way past them, and AI can only add caution.

Receipt before action: no durable record, no action. Each receipt is signed and chained.

One engine, many ways in: holds and undo apply when KAVACH runs the tool; when a host runs it, those stay the host's job.



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AI Usage & Innovation

Rules decide. AI reads meaning a keyword check misses, and can only add caution.

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AI CAPABILITY USED

Agentic workflow + calibrated risk + optional LLM semantic review

INNOVATION / DIFFERENTIATION

Argument lineage + consequence evidence + exact-action approval

AI ROLE

Risk scores support review. Default Qualcomm judge adds notes on escalations; explanations use templates.

INPUT & MODEL

User request + call + source labels; risk ensemble; two-stage Llama judge on Qualcomm.

QUALITY CHECK

Models cannot relax policy. Risk-zero ablation changed no suite verdicts. Optional rewrites are checked.

VALUE CREATED

Live inline test: 7/12 semantic misreadings caught vs 1/12 heuristic; safe work 11/12 in both.

EVIDENCE

- Rules and the intent gate carry the synthetic-suite decisions. The learned score is supporting evidence. The 24 semantic cases are hand-written; 7/12 in two clean live runs, 6/12 in a third with other jobs competing for the service.



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How KAVACH Compares

Each approach solves part of the problem. KAVACH combines the checks at the tool call.

Approach (example)	Decides at the tool call	Traces where values came from	Reads the real effect first	Hold or undo before it lands	Approval tied to the exact call	Signed, verifiable receipts
Text scanner (Infosys PromptGuard, RAI proxy)	—	—	—	—	—	—
Prompt + reasoning scanner (Meta LlamaFirewall)	Partly	—	—	—	—	—
Tool name + schema check (NVIDIA NeMo tool rails)	Yes	—	—	—	—	—
Pattern gate + hooks (Infosys AgentOS)	Yes	Partly	—	—	Partly	—
Policy gateway (AWS AgentCore Policy)	Yes	—	—	—	—	—
Least privilege (Progent)	Yes	—	—	—	Partly	—
Information flow (Google CaMeL)	Yes	Yes	—	—	Partly	—
Information flow (Microsoft FIDES)	Yes	Yes	—	—	—	—
KAVACH	Yes	Yes	Yes*	Yes*	Yes*	Yes

OUR USP: each approach above checks part of an action. KAVACH brings it all together at the tool call: where each value came from, what the call would really change, a person's approval for that exact call, a chance to stop it, and a signed record of what happened.

* SQL and files are measured in a sandbox; email and payments are predicted. Hold, undo and exact-call approval apply when KAVACH runs the tool. — = not a feature in the paper or docs we read, which doesn't prove it can't be added. Stronger elsewhere: CaMeL and FIDES enforce information flow by design, if the agent runs inside their planner; KAVACH's tracing is value matching, with known limits.



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Infosys AgentOS and KAVACH

AgentOS is Infosys's own agent runtime. We read and ran its code. KAVACH plugs into it; it does not replace it.

	Infosys AgentOS (built-in paths we read)	KAVACH
What it looks at	Hooks, tool and skill names, and patterns over commands and arguments	Where each value came from, and what this call would change
Injected email	PromptGuard and the RAI proxy paths don't see tool output; a custom hook could	Output labelled untrusted; an address taken from it is refused
Business limits	approval_rules are evaluated inside a tool the agent must choose to call, on context it supplies	Rules run outside the model; payment facts come from records (demo company today)
Approval card	Durable holds, with a stale-config check; card shows tool, command preview, skill, rule	Effect, each argument's origin, explanation; tied to the exact call
Undo	Holds pause a call; "rollback" means tool-code versions	Hold, recall, park or snapshot when KAVACH runs the tool
Audit log	JSONL and OpenTelemetry; no linked hash chain in the path we read	Hash chain, HMAC, Ed25519, signed checkpoint

Same suite, 282 test cases	Unsafe allowed	Safe work done	Sent to a person
AgentOS default gate	96.2%	100.0%	0.0%
AgentOS hardened gate	3.9%	14.3%	89.4%
KAVACH	**0.0%**	**95.2%**	**26.6%**

Hardening the AgentOS gate stops the injection attacks (0% success), but sends most actions to a person, which risks approval fatigue. On this suite KAVACH asks far less often, because it knows each call's origin and effect. Suite written by our team, seed 0; Infosys gate code loaded and run locally.

We plug in, we don't replace. AgentOS keeps routing, sandboxing, approval operations and RBAC, where it is ahead of us; KAVACH joins through its PreToolUse hooks and supplies the judgment. Tested through AgentOS's governance code with a stand-in agent; a full LLM-driven AgentOS chat isn't verified. There AgentOS runs the tool, so KAVACH's hold and undo don't apply.



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Future Enhancements, Scalability & Security

What works now, what comes next, and the safeguards already in place.

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NEXT ENHANCEMENTS

1. Add CI, load tests and backup / restore drills.
2. Complete wider paired attack evaluations; tune false escalations.
3. Validate reviewer usability and accessibility beyond the demo.

SCALABILITY

1. Single receipt-chain writer; Helm defaults to one replica.
2. Judge advises escalations; allowed calls avoid that model latency.
3. Prometheus / Grafana exist; alert delivery remains to be configured.

SECURITY & RESPONSIBLE AI

1. Signed caller identity and role checks when auth is configured.
2. Secrets broker; tool arguments omitted from telemetry.
3. Digest-bound approvals, kill switch and tamper-evident receipts.

NOW / NEXT

- Implemented: guarded workflows and recovery controls. Next: deployment validation. Demo company and bank are simulated; production readiness is unproven.



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DEMO



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